

Software Requirements Specification (SRS): Banking Management System

1. Project Overview

The Banking Management System is a comprehensive digital platform designed to facilitate secure financial services. It provides customers with core banking functionalities (transfers, bill payments, loan applications) and value-added features such as AI-driven spending insights and budget management. It also provides internal tools for employees and administrators to manage operations and risk.

2. Executive Summary

This system aims to modernize retail banking by providing a secure, intuitive, and data-driven interface. By integrating fraud detection and automated financial insights, the platform intends to increase user engagement and operational efficiency while maintaining high standards of security through multi-factor authentication (MFA).

3. Stakeholders

- **Customers:** Individual users managing personal finances.
- **Bank Employees:** Personnel responsible for verifying and approving loan applications.
- **System Administrators:** Staff responsible for monitoring system health, user access, and security logs.
- **Compliance/Audit Team:** Users responsible for monitoring suspicious activity reports.

4. Target Users

- **Retail Bank Customers:** Individuals seeking to manage deposits, transfers, and loans.
- **Operations Staff:** Employees performing manual underwriting for loans.
- **Super-Users (Admins):** Staff managing system configurations and user roles.

5. Functional Requirements

- **Account Management:** Users shall be able to create accounts, view balances, and manage profile settings.
- **Transaction Processing:** System shall support internal transfers, external payments, and bill pay.

- **Loan Management:** Users shall submit loan applications; Employees shall approve/reject based on set criteria.
- **Fraud Detection:** System shall flag transactions that deviate from historical patterns for review.
- **AI Financial Insights:** System shall analyze spending habits and provide monthly insights and budget recommendations.
- **Authentication:** System shall enforce MFA for all user sessions.
- **Audit Logging:** System shall log all critical actions (transfers, approvals, settings changes).

6. Non-Functional Requirements

- **Security:** Data must be encrypted at rest and in transit.
- **Availability:** The system shall maintain 99.9% uptime.
- **Scalability:** The architecture must support a 20% year-over-year growth in user accounts.
- **Compliance:** System must adhere to relevant regional banking regulations (e.g., GDPR, PCI-DSS).

7. User Stories

- *As a Customer,* I want to see a monthly breakdown of my spending so I can manage my budget better.
- *As a Customer,* I want to apply for a loan through the portal so I don't have to visit a branch.
- *As an Employee,* I want a dashboard showing pending loan applications so I can process them in order.
- *As an Admin,* I want to receive alerts on suspicious transactions to mitigate fraud risks.

8. Acceptance Criteria

- **MFA:** User cannot access dashboard without verifying secondary factor (OTP/App).
- **Loans:** Application status changes from 'Pending' to 'Approved' only after an employee triggers the approval action.
- **Transactions:** Funds are deducted from the sender and added to the receiver within 3 seconds of confirmation.

9. Business Rules

- Loan applications cannot be approved if the applicant has an existing default or a credit score below a defined threshold.
- Transactions exceeding \$5,000 require an extra layer of secondary authentication.
- All users must change their password every 90 days.

10. Assumptions

- All users have access to a mobile device for MFA.
- External credit bureaus provide the necessary data for loan eligibility.
- Internet connectivity is assumed for all platform operations.

11. Constraints

- Must comply with local data residency laws.
- Budgeting insights must be limited to account activity within the current system.

12. Risks

- ****False Positives:**** Fraud detection algorithms may block legitimate transactions.
- ****Security Breaches:**** High-value target for attackers; requires constant updates to security protocols.
- ****Regulatory Changes:**** Laws governing banking are subject to frequent change.

13. Out of Scope

- Physical hardware (ATMs, kiosks).
- Cryptocurrency or stock trading accounts.
- Integration with third-party global banking networks (SWIFT) in the MVP phase.

14. Open Questions

- What is the specific list of third-party vendors for credit checks?
- Should the AI insight model be retrained daily or weekly?
- What are the specific thresholds for "fraudulent" activity?